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Project Documentation

Project Type: Application Development Project

Project Name: INNOVESTIA

A Flutter Application for Startup and Investor Discovery

Submitted By
Shrushti Panditrao Deshmukh — 2501042

Under the Guidance of:
Prof. Nikita Tannu

Introduction

Innovestia is an Android-focused application that gives startups and investors a shared place to create profiles, discover the other account type, express interest, and continue initial discussions. Its purpose is to make early-stage startup-investor discovery and communication more organized. The current project is a client-server application: the Flutter client presents the mobile screens, a Node.js/Express REST API handles validation and access control, MongoDB stores application records through Mongoose, and startup pitch-deck PDF files are kept in a private local backend directory.

The implemented flow begins at a splash screen. At app startup, the client checks for a locally stored session token and asks the API to validate it. A user without a valid session reaches the welcome screen, then can register as a Startup or Investor or log in. Registration validates input, hashes the password on the server, creates a User and an initial Profile, and returns a signed JWT. The Flutter client stores the session with Flutter Secure Storage and opens the dashboard for the returned role. Authenticated API requests carry the JWT. From the role-aware dashboard, users can edit their profile, search and view opposite-role profiles, save profiles, and move into requests, messages, notifications, or profile management. An Investor can express interest in a Startup; a Startup can accept or reject a pending request. Eligible users can start a one-to-one conversation and exchange text messages. Startup users can upload, download, replace, or delete a PDF pitch deck. Logout clears local session storage.

The source does not implement OTP/email verification, a password-reset flow, push notification delivery, AI matching, meeting scheduling, payment/fund transfer, or analytics reports. Messaging refreshes by polling the API and is not a WebSocket service. Admin APIs support summary counts, a user list, and an account verification update, but the Admin screen only lists users; a verification button is not implemented in the inspected client. A successful code path is not the same as a verified live workflow, so the report distinguishes automated tests from end-to-end flows that remain unverified.

Technology Stack

Layer / technology

Use in the actual project	
Flutter and Dart	Android application UI, navigation, stateful screens, and client logic
Material widgets	Theme, form controls, navigation bar, cards, dialogs, and lists
http package	JSON REST requests and binary PDF transfer
flutter_secure_storage	Stores the client-side JWT and serialized user session
file_picker	Selects a pitch-deck PDF and offers a save destination for downloads
Node.js	Backend runtime using JavaScript ECMAScript modules
Express.js	HTTP API, route mounting, middleware, JSON parsing, CORS, and health endpoints
MongoDB	Document database used for application data
Mongoose	Model schemas, references, validation, indexes, and database operations
jsonwebtoken	Signs and verifies bearer session tokens; default token lifetime is seven days
bcryptjs	Hashes passwords on registration and compares hashes at login; cost factor 12
express-rate-limit	Applies global API limits and a separate login/registration limit
dotenv	Loads backend configuration from backend/.env
cors	Applies origin handling for browser-based clients

The app API URL is configured with API_BASE_URL. Its defaults are http://10.0.2.2:8081/api for Android Emulator and http://localhost:8081/api for desktop/web. The database connection requires the URI to select innovestia_app; Windows explicitly opens the existing on-disk spelling Innovestia_app. The project does not use Firebase Authentication, FCM, Cloudinary, FastAPI, or a Python machine-learning service. The README mentions optional email-provider configuration and says remote storage is needed, but the inspected backend has no email/OTP sender and implements local pitch-deck storage.

Functionalities / Modules

Module	Implemented responsibility	Current status
Authentication	Startup/Investor registration, login, JWT session restoration, local logout, and retry cooldown	Implemented in source. Automated tests cover client behavior and throttling; full live signup/login is not covered by the automated suite.
Profile management	Read/update one profile per user; role-specific fields are submitted by the Flutter profile form	Connected to API and MongoDB through Profile model
Discovery and saved profiles	Search opposite account type; view profile details; save/remove profiles	Implemented through authenticated profile and SavedProfile endpoints
Investor interest and requests	Investor creates a request; Startup reviews and accepts/rejects pending request	Implemented in backend and client. No investment transaction is performed. WITHDRAWN exists in schema but is not accepted by update controller.

Messaging	
Search opposite-role users, create/reuse pair conversation, send text, load messages and unread counts	
Implemented using API polling; not real-time WebSocket messaging	
Notifications	
Persist and list in-app notification records; mark one or all read	
Implemented as database/API feature; no mobile push provider is present	
Pitch deck	
Startup PDF upload/delete; owner or Investor download of startup PDF	
Implemented with local filesystem storage; no cloud object storage	
Admin	
Summary counts, user listing, server endpoint to mark a user verified	
API implemented; client lists users but does not expose a verify action	
Search and matching	
Text and field filters on the opposite-role profile list	
Search implemented; no AI/automated recommendation model	
Meetings, payments, reports, analytics	
Not found as implemented screens, API routes, controllers, or models	
Not implemented in current source	

User Roles

Role	Source-backed permissions and behavior
Startup	Register/login; edit startup profile; discover and save investor profiles; receive and decide pending investment-interest requests; upload, download, or delete own pitch deck; message eligible Investors; read notifications
Investor	Register/login; edit investor profile; discover and save startup profiles; express interest in a Startup; list own requests; download an available Startup pitch deck; message eligible Startups; read notifications

Admin

Role accepted by User schema and protected admin routes; view summary counts, list up to 100 users, and call the verification API. Admin accounts cannot be created through public registration, which accepts only Startup and Investor.

The client routes an authenticated Admin role to the shared role-aware dashboard. Admin user verification is available on the backend but no corresponding verification control was found in the Flutter user-list screen. The schema's isVerified flag is an administrative status, not evidence of OTP/email verification.

Number of User Interfaces

The source contains 12 main screen widgets/routes, with shared role-specific dashboard content, plus a profile-detail bottom sheet. Saved Profiles is a mode of the Discovery screen rather than a separate route.

Group

Screens/components found in source

Shared entry

Splash, Welcome, Login, Registration

Startup and Investor

Shared Dashboard screen (role-specific content), Discovery/Saved Profiles, Profile Detail bottom sheet, Investment Requests, Messages, Conversation, Notifications, Profile/Pitch Deck management

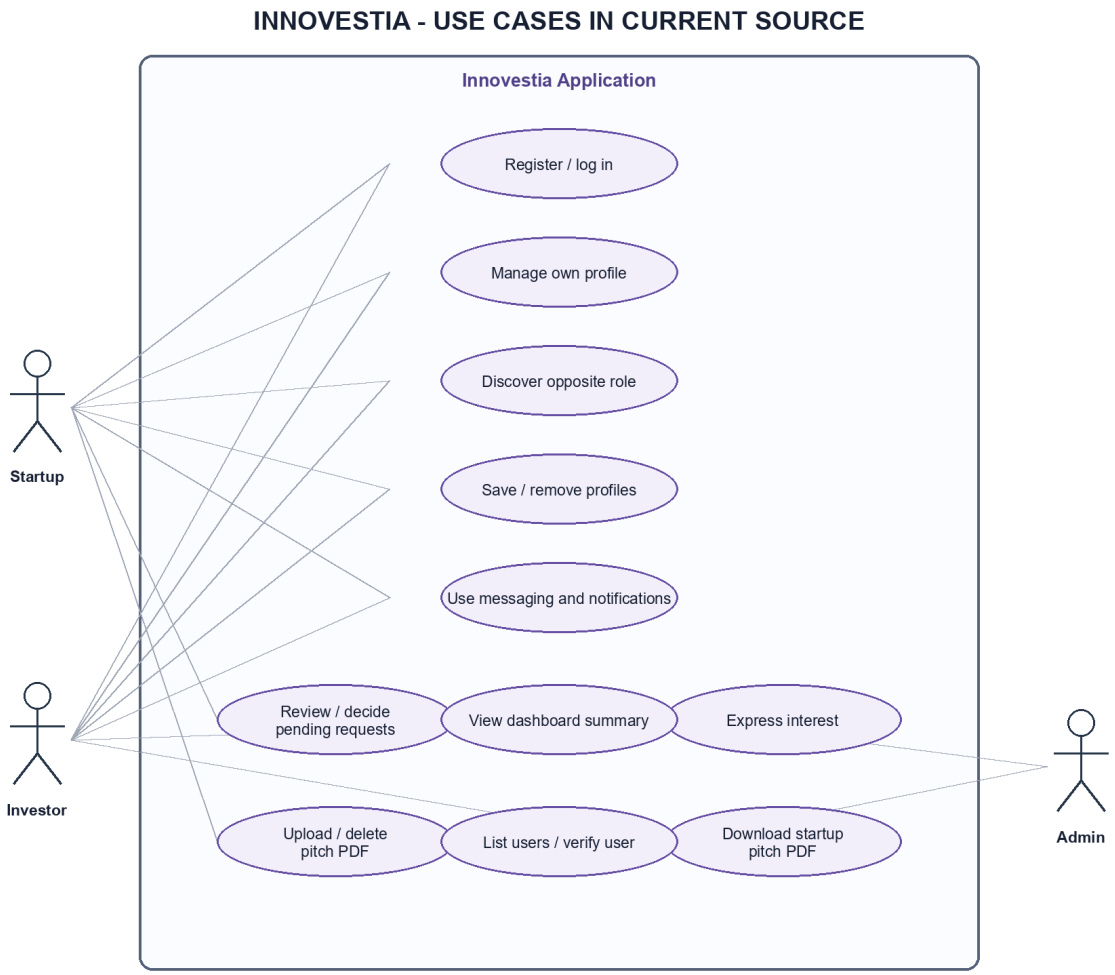
Admin

Shared Dashboard screen with Admin summary content; Admin User Management screen

The dashboard bottom navigation includes Home, Discover, Messages, Alerts, and Profile. The same screen class is reused for Startup, Investor, and Admin dashboard roles. No separate OTP, meeting, AI recommendation, report management, or analytics screen exists in the inspected Flutter source.

Use-Case Diagram

The diagram shows only source-backed use cases. Admin verification is marked as an API capability because the current Flutter user list does not expose that action.

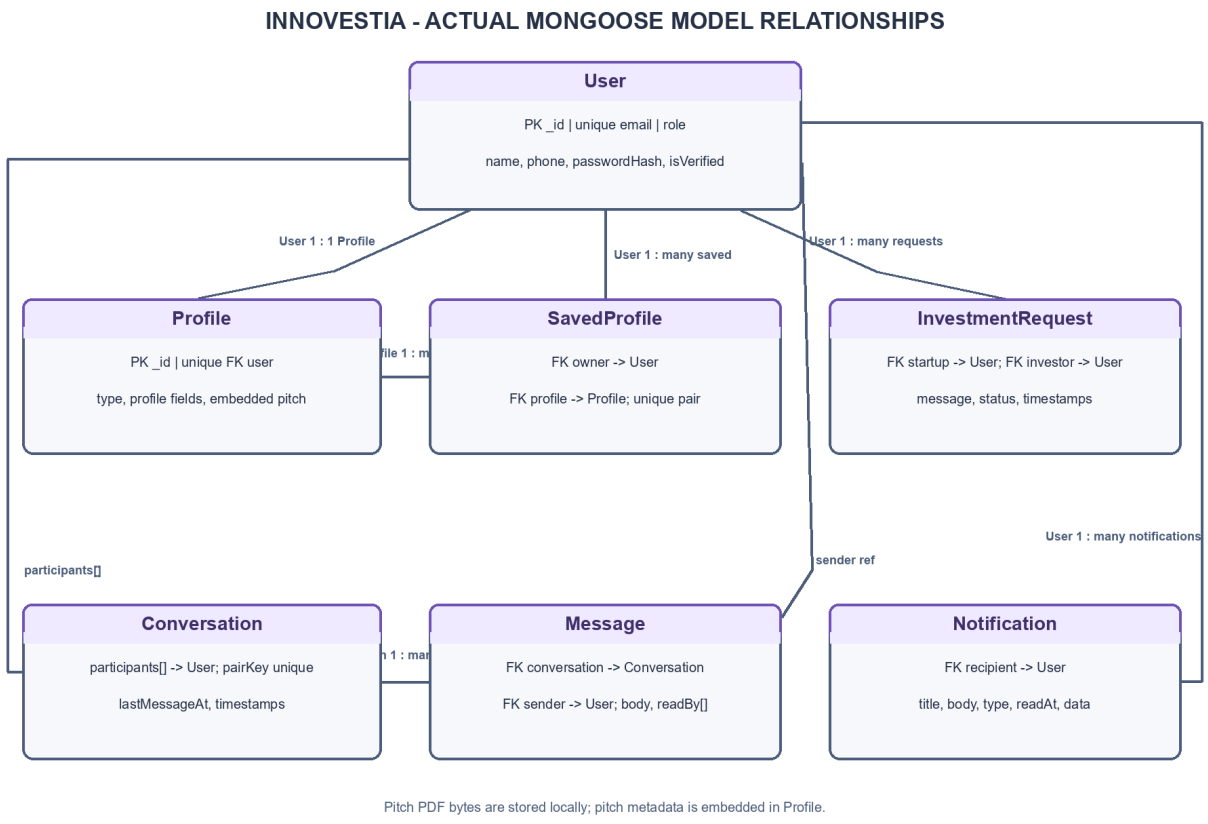


Use-case diagram based on implemented Flutter screens and API operations

Figure 1. Innovestia use cases verified from current source.

ER Diagram

The model diagram reflects the seven Mongoose models in backend/src/models. Profile is a single model with a type field, and Admin is a User role rather than a separate collection. Pitch metadata is embedded in Profile; PDF bytes are stored on the backend filesystem.



ER diagram of actual Mongoose models and references

Figure 2. Actual User, Profile, SavedProfile, InvestmentRequest, Conversation, Message, and Notification relationships.

Table Structure / Data Dictionary

Mongoose supplies each document's _id and timestamps where the schema enables timestamps. Collections below are the models used by source. These are document references and indexes, not SQL foreign-key constraints.

1. Users Collection

Field Name	Data Type	Key / Validation	Description
_id	ObjectId	Primary identifier (Mongoose)	User identifier
name	String	Required, trimmed, max 100	Account/display name
email	String	Required, unique, lowercase, trimmed	Login identity
phone	String	Optional, trimmed	

Contact number

passwordHash

String

Required; select:false

bcryptjs hash; omitted by default from queries

role

String

Required enum STARTUP, INVESTOR, ADMIN

Account role and authorization

isVerified

Boolean

Default false

Admin verification flag

resetHash

String

Optional

Reset-related field; no reset flow found

resetExpiresAt

Date

Optional

Reset-related field; no reset flow found

createdAt, updatedAt

Date

Mongoose timestamps

Record creation/update time

2. Profiles Collection

Field Name	Data Type	Key / Validation	Description
_id	ObjectId	Primary identifier	Profile identifier
user	ObjectId	Required, unique reference to User	Profile owner; one profile per user

type

String

Required enum STARTUP, INVESTOR

Profile category

name

String

Required, trimmed, max 100

Profile name

location

String

Optional, max 120

Location and discovery data

industry

String

Optional, max 120

Industry and discovery data

businessStage

String

Optional, max 120

Business stage and discovery data

description

String

Optional, max 2,000

Profile description

website

String

Optional, max 500; controller requires HTTP/HTTPS when provided

Website URL

fundingRequired

Number

Optional; controller requires non-negative

Startup funding need

fundingRaised

Number

Optional; controller requires non-negative

Startup funding raised

investmentRangeMin

Number

Optional; controller requires non-negative

Investor range minimum

investmentRangeMax

Number

Optional; controller requires non-negative

Investor range maximum

preferences

Array of String

Optional

Profile preferences; used in discovery search

pitch.url, pitch.filename

String

Optional embedded fields

Local deck key and filename

pitch.title, pitch.summary, pitch.description

String

Optional embedded fields

Pitch-deck metadata

pitch.size

Number

Optional embedded field			
PDF byte count			
pitch.updatedAt			
Date			
Optional embedded field			
Upload time			
createdAt, updatedAt			
Date			
Mongoose timestamps			
Profile creation/update time			

Profile has a compound index on type, industry, location, and businessStage. The controller requires a profile name and checks investmentRangeMin does not exceed investmentRangeMax when both are supplied. Funding units/currency are not defined in the schema.

3. SavedProfiles Collection

Field Name	Data Type	Key / Validation	Description
_id	ObjectId	Primary identifier	Saved record
owner	ObjectId	Required reference to User	

Account that saved the profile

profile

ObjectId

Required reference to Profile

Saved profile

owner + profile

Compound index

Unique

Prevents duplicate saves by one owner

createdAt, updatedAt

Date

Mongoose timestamps

Save record times

4. InvestmentRequests Collection

Field Name

Data Type

Key / Validation

Description

_id

ObjectId

Primary identifier

Request identifier

startup

ObjectId

Required reference to User

Startup receiving interest

investor

ObjectId

Required reference to User

Investor sending interest

message

String

Optional, max 1,000

Investor’s message

status

String

Enum PENDING, ACCEPTED, REJECTED, WITHDRAWN; default PENDING

Request state

startup + investor + status

Compound index

Non-unique

Request query index

createdAt, updatedAt

Date

Mongoose timestamps

Request times

The controller prevents a second pending request through application logic, not a unique index. Request update accepts only ACCEPTED or REJECTED for a pending request, even though WITHDRAWN is an allowed schema value. There is no amount field in this model.

5. Conversations Collection

Field Name	Data Type	Key / Validation	Description
_id	ObjectId	Primary identifier	Conversation identifier
participants	Array of ObjectId	References to User; each array entry required	Conversation participants; API creates a pair of opposite-role users
pairKey	String	Optional sparse unique index	Sorted pair key used to reuse a conversation
lastMessageAt	Date	Optional	Conversation ordering
participants + lastMessageAt	Compound index		

Non-unique

Conversation lookup/order index

createdAt, updatedAt

Date

Mongoose timestamps

Conversation times

6. Messages Collection

Field Name

Data Type

Key / Validation

Description

_id

ObjectId

Primary identifier

Message identifier

conversation

ObjectId

Required, indexed reference to Conversation

Parent conversation

sender

ObjectId

Required reference to User

Sending account

body

String

Required, max 4,000

Text message

readBy

Array of ObjectId

References to User

Participants recorded as having read it

createdAt, updatedAt

Date

Mongoose timestamps

Message times

The Message model does not contain receiverId, attachment URL, or messageType. Conversation participants determine eligible members.

7. Notifications Collection

Field Name

Data Type

Key / Validation

Description

_id

ObjectId

Primary identifier

Notification identifier

recipient

ObjectId

Required, indexed reference to User

Recipient

title

String

Required

Notification title

body

String

Required

Notification body

type

String

Optional

Examples created by source include MESSAGE and INVESTMENT_REQUEST

readAt

Date

Optional

Read time; missing/null is counted as unread

data

Mixed

Optional

Related request/conversation metadata

createdAt, updatedAt

Date

Mongoose timestamps

Notification times

Sample Input and Output Screens

The project contains no valid application screen captures. A file named flutter_01.png is not a readable image, and the pictures embedded in the supplied Word document are old diagrams, not screenshots of the current app. Add real emulator/device captures in the marked spaces below; do not treat these captions as evidence of runtime testing.

Figure

Screenshot placeholder

Figure 3

[Insert current Welcome and Login screen capture]

Figure 4

[Insert current Registration screen with role selection]

Figure 5

[Insert Startup dashboard screen capture]

Figure 6

[Insert Investor dashboard and Discover list capture]

Figure 7

[Insert profile detail, saved profile, and interest controls]

Figure 8

[Insert Investment Requests screen and Startup decision menu]

Figure 9

[Insert Messages conversation and Notifications screen]

Figure 10

[Insert Profile editor and Startup pitch-deck controls]

Figure 11

[Insert Admin dashboard and User Management screen]

Example API Input / Output

The registration screen submits name, email, phone, password, and role; the API accepts STARTUP or INVESTOR, hashes the password, creates the user/profile, and responds with a token and public user object. The token is omitted here and must never be copied from a live session. Invalid registration data returns HTTP 422, duplicate email returns 409, and invalid login returns 401. Repeated auth attempts can return 429 with Retry-After: 15. Automated tests cover these selected responses; full live account creation is not verified by the test suite.

User Manual

The instructions below describe the UI and API behavior found in source, not a claim that every workflow has been tested on a physical device.

1. Launch

Install the project APK or run the Android app with Flutter tools.

Start the backend and MongoDB using the app database configured for the project.

Set the API URL to a host reachable from the device. The Android Emulator default is http://10.0.2.2:8081/api; a physical phone needs a reachable LAN address or HTTPS API.

Launch Innovestia. The splash screen transitions to Welcome or to the role dashboard after session validation.

2. Register or Log In

On Welcome, choose Create Account or Login.

Registration asks for name, email, phone, password, and Startup/Investor role. Client validation requires a valid email and a password containing at least eight characters, a letter, and a number.

On successful registration, the API creates a user and starter profile and returns a JWT; the app stores the session and opens the role dashboard.

Login uses email/password. The server verifies the password hash and returns the user's role and token.

There is no OTP/email verification screen or password-reset workflow in current source.

If the server returns 429, wait for the displayed 15-second cooldown.

3. Startup Workflow

Open Profile to edit name, location, industry, business stage, description, website, funding required, and funding raised.

Save the profile. Website must be HTTP/HTTPS and numeric values must be non-negative.

Open Discover Investors, search or browse, open a result, and save/remove profiles.

Review Investment Requests. For a pending request, choose Accept or Decline.

Use Messages to search for an eligible Investor, start/open a conversation, and send text.

In Profile, choose a PDF, enter a deck title and optional summary/description, then upload. Server limit is 10 MB. Existing deck can be downloaded, replaced, or deleted.

Open Alerts to read notifications; tap one to navigate to related request or conversation.

Use Profile → Logout and confirm to clear the local session.

4. Investor Workflow

Open Profile to edit name, location, industry, description, website, and investment range.

Open Discover Startups to search profiles and view details.

Save/remove a profile, download an available startup pitch PDF, or choose Express Interest.

Open Investment Requests to view submitted request status.

Use Messages and Alerts to continue the discussion and read updates.

5. Admin Workflow

Sign in using an Admin account provisioned outside the public registration form.

Review the platform summary and open User Management.

The Flutter screen lists users; it has no verification action. The backend exposes a protected verification endpoint for an authorized Admin API client.

6. Common Errors

Error	Meaning / user action
Could not connect / timeout	Check server availability, network route, and API_BASE_URL
Invalid email or password	Re-enter account credentials
Email already registered	Use Login for the existing account

Too many attempts (429)

Wait 15 seconds before retry

Invalid website

Enter full HTTP or HTTPS URL

PDF upload rejected

Use Startup account, valid PDF under 10 MB, and provide a deck title

Access denied

The endpoint requires an eligible role or resource owner

Testing

A previously recorded project run passed the 12 tests in Flutter's test/widget_test.dart. The backend suite in backend/test/auth-rate-limit.test.js passed 2 tests during this analysis. A fresh Flutter test run was attempted while preparing this report, but the test runner stalled without producing results; its status could not be reconfirmed during this update.

ID

Area

Test scenario

Expected result

Actual result

T-01

API client

API URL normalization and ngrok/local headers

Correct path/header behavior

Pass

T-02

API client

Preserve server error response

Client exposes backend message

Pass

T-03

API client

Handle 429 retry-after header

Expose 15-second retry value

Pass

T-04

Login UI

Simulate 429 and advance cooldown

Submit disabled then re-enabled

Pass

T-05

Registration UI

Simulate 429 cooldown

Submit disabled during wait

Pass

T-06

Validators

Email and password validation

Valid values accepted, invalid rejected

Pass

T-07

Roles

API role serialization and dashboard routing

Supported roles map to expected dashboard

Pass

T-08

Backend throttle

11 registration attempts in test server

11th returns 429 and Retry-After 15

Pass

T-09

Profile validation

Invalid website passed to controller

422 before database write

Pass

T-10

Live account persistence

Register/login against MongoDB

User/profile persist and authenticate

Not covered / Not Verified

T-11

Two-account workflow

Interest decision, message, notification

Records persist and role restrictions hold

Not covered / Not Verified

T-12

PDF round trip

Upload/download/delete using running backend

File and metadata behave correctly

Not covered / Not Verified

T-13

Device acceptance

Install release on physical Android and exercise flows

Main workflows operate on target device

Not Tested in this report

These test results establish only the scenarios above. They do not prove production deployment, physical-device acceptance, email verification, push delivery, or complete two-account functionality.

Conclusion

The current Innovestia project is a Flutter Android client connected to a Node.js/Express API and MongoDB/Mongoose backend. It implements Startup and Investor account flows, profiles, opposite-role discovery, saved profiles, investor interest requests with Startup decisions, polling-based text conversations, database notifications, and local PDF pitch-deck management. The backend also includes an Admin summary, user list, and verification endpoint. Password hashing, JWT-based access, role/ownership checks, validation, and rate limiting are present in code. The report deliberately does not claim that OTP, email, push messaging, AI recommendations, meetings, payments, or reporting work. Automated tests pass for selected client/API and backend validation cases, while full live workflows and device acceptance remain unverified.

Future Scope

The following capabilities are possible extensions and are not described as current features:

Implement verified email/OTP and password-reset flows with a configured provider.

Add server-side session revocation or refresh-token rotation.

Add an Admin verification action to the Flutter interface and define secure Admin provisioning.

Use managed object storage, access controls, backup, and file scanning for pitch decks.

Add push delivery or event-driven messaging if product requirements call for it.

Add the WITHDRAWN request action if startup withdrawal is required.

Add investment pipeline analytics, meetings, reports, or recommendations only with corresponding data models, APIs, UI, and tests.

Expand integration tests for real database persistence, two-account workflows, file storage, and Android devices.

Configure a unique Android application ID, production signing, HTTPS deployment, monitoring, backups, and production secret management.

Accuracy Notes and References

The supplied template’s original technical claims were not carried over unless confirmed in current code. No real application screenshots were found. Admin account creation is not exposed through public signup. The current source uses local pitch file storage; this differs from a README note about remote storage configuration. No credentials or secret environment values are included.

Technical references: Flutter documentation (https://docs.flutter.dev/), Dart documentation (https://dart.dev/), Node.js documentation (https://nodejs.org/docs/), Express documentation (https://expressjs.com/), MongoDB documentation (https://www.mongodb.com/docs/), and Mongoose documentation (https://mongoosejs.com/docs/). Project source reviewed: lib/, backend/src/, test/, backend/test/, pubspec.yaml, backend/package.json, Android configuration, and the supplied README.